

Assignment 1C

Bagging And Boosting

→ Ensemble Learning: Ensemble learning basically means a technique where multiple models of machine learning are combined and trained to solve the same problem. The most common ensemble techniques include:-

- (1) Bagging
- (2) Boosting

(1) Bagging :- Bagging is an ensemble learning technique that reduces model variance by training multiple independent models on different bootstrap samples of same dataset and aggregating their outputs.

Working of Bagging

- (i) From the original training ^{data} test, several new datasets are created using random sampling with replacement
- (ii) Each dataset is used to train a separate AI model (such as decision tree)
- (iii) All models learn independently of each other.
- (iv) The final output is obtained by combining predictions
 - classification :- majority voting
 - Regression :- averaging

Objectives of Bagging:-

- Reduce variance in AI models
- Improve stability and reliability
- Avoid Overfitting

Advantages of Bagging

- Improves performance of unstable learners
- Reduces overfitting
- Models can be trained in parallel
- Robust against noise

Disadvantages of Bagging

- Does not reduce Bias
- Requires more memory and computation
- Less effective for simple AI models

Examples:-

- Random Forest

↳ multiple decision trees act as intelligent agents.

↳ used in image recognition and medical diagnosis.

② **Boosting**:- Boosting is an ensemble learning techniques in AI where models are trained sequentially, and each new model focuses more on the errors made by previous models, thereby improving the intelligence of the system.

Working of Boosting:

- ① All training samples are initially assigned equal importance
- ② A weak AI learner is trained on the dataset
- ③ Incorrectly classified samples are given higher priority
- ④ A new model is trained focusing on difficult cases

- ⑤ This process continues for several iterations
- ⑥ Final decision is made using weighted combination of all models.

Objectives of Boosting

- ↳ Reduce bias in AI systems
- ↳ Increase learning accuracy
- ↳ Improve handling of complex patterns

Advantages of Boosting

- ↳ Convert weak AI learners into strong learners
- ↳ high accuracy in prediction
- ↳ effective for complex and non-linear problems

Disadvantages of Boosting

- ↳ Highly sensitive to noise and outliers
- ↳ Sequential training increases time complexity
- ↳ Risk of overfitting noisy data

Examples of Boosting

- ↳ AdaBoost
- ↳ Gradient Boosting
- ↳ XGBoost
- ↳ LightGBM
- ↳ CatBoost

Q. Comparison of Bagging and Boosting in AI

feature	Bagging	Boosting
Learning style	parallel learning	sequential learning
Model dependency	Independent AI agents	Dependent AI agents
Main goal	Reduce variance	Reduce Bias
Data handling	bootstrap sampling	weighted data samples
Noise sensitivity	Low	High
overfitting	Reduces	may increase

Applications in Artificial Intelligence

• Bagging Applications

- ↳ Medical diagnosis systems
- ↳ Spam detection
- ↳ Image classification

• Boosting Application

- ↳ Face Recognition
- ↳ Speech Recognition
- ↳ Fraud detection
- ↳ Recommendation systems.

In ~~art~~ Artificial Intelligence, Bagging & Boosting are essential ensemble learning techniques that enhance the decision-making capability of intelligent systems.

② Machine Learning model

Machine Learning model is computational program that learns patterns from dataset and makes decision or prediction on new unseen data. It is created by training a machine learning algorithm on dataset and optimizing it to minimize errors.

Machine learning models can be broadly categorized into four primary paradigm based on the nature of data and learning data.

- ① Supervised learning models
- ② Unsupervised learning models
- ③ Semi-Supervised learning Models
- ④ Reinforcement learning models

① Supervised learning model:-

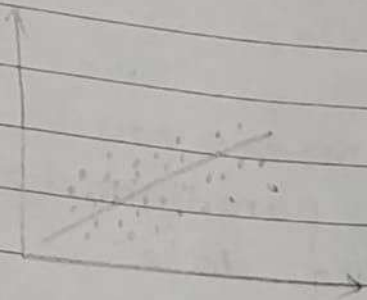
Supervised learning models learn from labeled data where each input has known output. The goal is to map input feature to the correct target value using mathematical model.

A ~~A~~ Regression :- Regression models predict continuous

- numerical value rather than categories. Some of its algorithms are
- (a) Linear Regression
 - (b) Polynomial Regression
 - (c) Decision Tree Regression
 - (d) Random Forest Regression
 - (e) Support Vector Regression

Linear Regression:-

Linear Regression analysis is used to create a model that describes the relationships between a dependent variable and one more independent variable.



For eg:- we want to predict a student's exam score based on how many hours they studied. We observe that as students study more hours, their scores go up. In the example of predicting exam score based on hours studied here,

- Independent Variable (Input):- Hours studied because it's the factor we control or observe
- Dependent variable (output):- Exam score because it depends on how many hours were studied

Advantages

- ① Linear Regression is computationally efficient and can handle large datasets effectively.
- ② It can be trained on large dataset making it suitable for real-time application.
- ③ Linear Regression is relatively robust to ^{outlier} compared to other machine learning algorithm.

Limitation

- ① Linear regression is susceptible to both overfitting and underfitting.
- ② It provide limited explanatory power for complex relationship between variable.
- ③ It can't perform well if the relationship is not linear.

⑧ Classification:- classification models assign input data to predefined categories. Some of its algorithm are:-

- a) Support vector machine (svm)
- b) Decision Tree
- c) Random forest
- d) Naïve Bayes
- e) K-Nearest Neighbour (KNN)

a) Support Vector Machine (svm):- It is powerful supervised machine learning algorithm primary used for classification problems, but they can also be applied to regression. It tries to find the best boundary known as hyperplane that

Separates different classes in data. It is useful when you want to do binary classification like spam vs not spam or cat vs dog.

Advantages

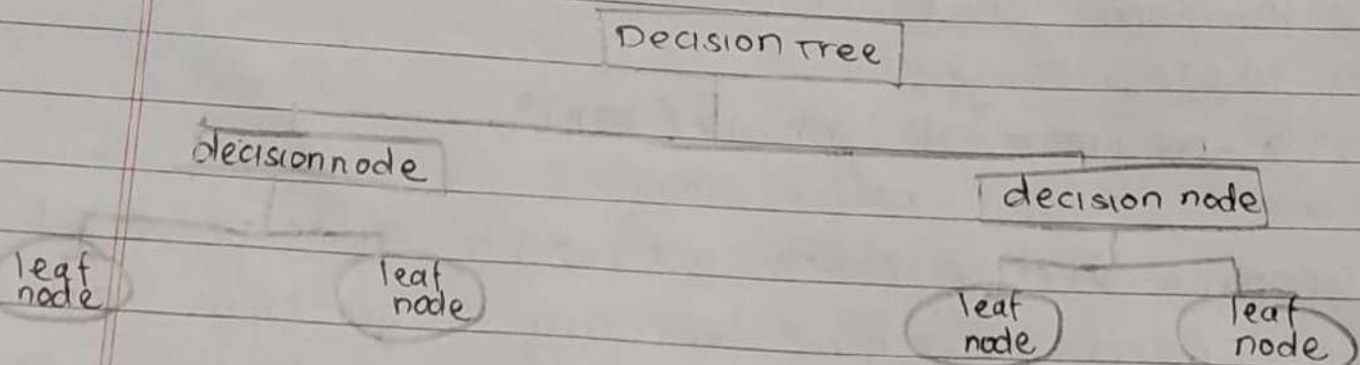
- ① High - dimension performance
- ② Non-Linear capability
- ③ Outlier Resilience
- ④ Binary and multiclass support
- ⑤ Memory efficiency

Disadvantages

- ① Slow training
- ② Parameter tuning difficulty
- ③ Limited Interpretability.

(b) Decision Tree

A decision tree helps us to make decision by mapping out different choices & their possible outcomes. It is used in machine learning for task, like classification & prediction.



Advantages

- ① Easy to understand
- ② Versatility
- ③ Handle non-linear relationships
- ④ Handles Missing data.

Disadvantages

- ① Overfitting
- ② Instability
- ③ Bias toward feature with many categories
- ④ Difficulty in capture complex instruction

③ Random Forest:-

Random forest algorithm is an ensemble learning method for classification and regression task consisting of many decision trees (to 'forest') to improve prediction accuracy.

Advantages:

- ① Random forest provide very accurate prediction even with large dataset
- ② It can handle missing data well without compromising with accuracy.
- ③ It doesn't require normalization or standardization on dataset

- ④ When we combine multiple decision tree it reduces the risk of overfitting problem.

Limitation.

- ① It can be computationally expensive especially with large number of tree.
- ② It's hard to interpret the model compared to simpler models like decision trees.

② Naive Bayes:

It is machine learning classification algorithm that predict the category of data point using probability. It perform well in many real-world application such as spam filtering, document categorization and sentiment analysis.

Advantages:

- ① Easy to implement & computationally efficient
- ② Effective in case with a large number of features
- ③ perform well even with limited training data.
- ④ It perform well in the presence of categorized feature.

Disadvantages

- ① Can be influenced by irrelevant attributes
- ② May assign zero probability to unseen events, leading to poor generalization.

② k-nearest Neighbour (KNN) algorithm

It is simplest yet powerful machine learning ~~Algo~~ algorithm used for classification and regression task. It works by finding the "k" nearest neighbour of a new data point and classifying or predicting its value based on majority or average of those neighbours.

Advantages:

- ① Simple to use
- ② No training step
- ③ Few parameter
- ④ Versatility

Disadvantages:-

- ① Slow with large data
- ② Struggle with many features
- ③ Can overfit.

Unsupervised learning

Unsupervised learning models work with unlabeled data, discovering hidden patterns, or structures without predefined outputs.

Clustering: Group similar data into cluster based on feature similarity. Some of its algorithm are:-

- ① K-Mean
- ② Hierarchical clustering.

① K-Mean :-

K-Mean clustering group similar data point into cluster without needing labeled data. It is used to uncover hidden pattern when the goal is to organize data based on similarity.

Limitation

- ① choosing the right no of cluster (k)
- ② sensitive to initial centroids.
- ③ Non-spherical cluster
- ④ Outliers.

② Hierarchical clustering

It is an unsupervised learning methods used to group similar data point into clusters based on their distance similarity.

Dimensionality Reduction:- Reduction high-dimensional data while retaining important information for analysis or visualization. Some of the technique are :-

① PCA (Principal component Analysis) :- Project data into principal component to reduce dimension.

② LDA (Linear discriminant Analysis) : Maximize class separability while reducing dimensionality.

Advantages of supervised learning

- (i) Simplicity & clarity
- (ii) High accuracy
- (iii) Versatility
- (iv) Generalization
- (v) wide application

Disadvantages of supervised learning

- ① Required labeled data
- ② Bias from data
- ③ Overfitting risk
- ④ Limited adaptability.

Advantages of Unsupervised Learning

- (a) No need for labeled data
- (b) Discovers hidden patterns
- (c) Handles complex and large datasets
- (d) Useful for anomaly detection.

Challenges of Unsupervised Learning

- (a) Noisy data
- (b) overfitting risks
- (c) limited Guidance
- (d) cluster interpretability.

③ K-Mean (Numerical)

Question

$A_1(2,10), A_2(2,5), A_3(8,4), B_1(5,8), B_2(7,5), B_3(6,4)$
 $C_1(1,2), C_2(4,9)$

Initial centroid (Iteration 1)

$A = (2,10)$

$B = (5,8)$

$C = (1,2)$

Point	X	Y	disA	disB	disC	cluster	new cluster
A1	2	10	0.00	8.606	8.062		A
A2	2	5	5.00	4.243	3.162		C
A3	8	4	8.485	5.00	7.280		B
B1	5	8	3.606	0.000	7.211		B
B2	7	5	7.071	3.60	6.708		B
B3	6	4	7.211	4.123	5.385		B
C1	1	2	8.062	7.211	0.00		C
C2	4	9	2.236	1.414	7.616		B

Recompute centroid

$$A = \text{mean}(\{(2,10)\}) = (2.0, 10.0)$$

$$B = \text{mean}(\{(8,4), (5,8), (7,5), (6,4), (4,9)\}) = (6.0, 6.0)$$

$$C = \text{mean}(\{(2,5), (1,2)\}) = (1.5, 3.5)$$

from distance formula $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ and

$$\text{centroid from} = \frac{\sum X}{\sum 1}, \frac{\sum Y}{\sum 1}$$

Point	x	y	disA	disB	disC	cluster	newcluster
A1	2	10	0.00	5.658	6.519	A	A
A2	2	5	4.6	4.123	1.581	C	C
A3	8	4	8.485	2.828	6.519	B	B
B1	5	8	3.605	2.236	5.700	B	B
B2	7	5	7.071	1.414	5.700	B	B
B3	6	4	7.211	2.00	4.527	B	B
C1	1	2	2.062	6.403	1.581	C	C
C2	4	9	2.236	3.605	6.041	B	A

Recompute centroid

$$A = (\{(2,10), (4,9)\}) = (3, 9.5)$$

$$B = (\{(8,4), (5,8), (7,5), (6,4), (4,9)\}) = (6.5, 5.25)$$

$$C = (\{(2,5), (1,2)\}) = (1.5, 3.5)$$

Point	x	y	disA	disB	disC	cluster	newcluster
A1	2	10	1.11	6.54	6.52	A	A
A2	2	5	4.60	4.51	1.58	C	C
A3	8	4	7.43	1.95	6.52	B	B
B1	5	8	2.50	3.23	5.70	B	A
B2	7	5	6.02	0.56	5.70	B	B
B3	6	4	6.26	1.35	4.53	B	B
C1	1	2	7.76	6.39	1.58	C	C
C2	4	9	1.12	4.51	6.04	A	A

New centroid

$$A = (\{(2,10), (5,8), (4,9)\}) = \left(\frac{2+5+4}{3}, \frac{10+8+9}{3}\right) = (3.66, 9)$$

$$B = (\{(8,4), (7,5), (6,4)\}) = \left(\frac{8+7+6}{3}, \frac{4+5+4}{3}\right) = (7, 4.33)$$

$$C = (\{(2,5), (1,2)\}) = \left(\frac{2+1}{2}, \frac{5+2}{2}\right) = (1.5, 3.5)$$

Point	x	y	disA	disB	disC	cluster	newcluster
A1	2	10	1.94	7.56	6.52	A	A
A2	2	5	4.83	5.04	1.58	C	C
A3	8	4	6.62	1.05	6.52	B	B
B1	5	8	1.67	4.18	5.70	A	A
B2	7	5	5.21	0.67	5.70	B	B
B3	6	4	5.52	1.05	4.53	B	B
C1	1	2	7.49	6.44	1.58	C	C
C2	4	9	0.33	8.55	6.04	A	A

Since the cluster and new cluster are equal
 It is the optional solution.

where

$$A = \{(3, 6.6, 9)\}$$

$$B = \{(7, 4, 3.3)\}$$

$$C = \{(1.5, 3.5)\}$$